

Case Study: Payment Exception & Reconciliation Workflow Redesign

Scope note (read this first): This project is a requirements-and-design-phase deliverable. No SQL was executed, no mock dataset was built, and no dashboard exists. Every claim below describes what the design specifies or is intended to achieve — none of it is a measured or tested result. This is a deliberate scope boundary, not an unfinished project; the value of this exercise is in the rigor of the requirements artifacts themselves.

The problem. Operations staff were manually reconciling transactions across three disconnected sources — an internal ledger, a payment gateway feed, and a clearing-house settlement file — using CSV exports and Excel lookups. Every mismatch, however trivial, went through the same undifferentiated manual investigation, and no resolution path left a standardized record of why it was resolved the way it was.

The conflict. This created a direct structural tension between two stakeholders. The Finance Controller needed exceptions cleared quickly enough to close the books on schedule, with minimal unresolved value sitting unexplained on the balance sheet. The Compliance/Audit Lead needed every exception properly investigated and logged before resolution — because an exception cleared too quickly, with no audit trail, might be hiding a genuine error or fraud rather than routine settlement noise. Under the existing manual process, neither stakeholder was well served: resolution was too slow and inconsistent for Finance, and too undocumented for Compliance to defend under audit sampling.

The resolution approach. Rather than choosing a single uniform speed for all exceptions, the design specifies a three-way matching framework that separates transactions by *category of risk*, not just by deadline:

- Transactions that match exactly (FR-001) or fall within a deliberately conservative, explicitly pragmatic 0.05% tolerance baseline (FR-002) are auto-resolved instantly, with no manual touch — satisfying Finance's need for throughput on the majority of routine volume.
- Everything else becomes a genuine Exception (FR-003), which is never auto-resolved and is instead aged into escalating priority buckets (FR-004) — 0–24h, 24–72h, and >72h — ensuring the longest-unresolved, highest-risk breaks receive escalating attention rather than waiting in an undifferentiated queue.
- Every resolution path — automated or manual — carries a mandatory, structured audit log (FR-005, FR-006), closing the gap that made the original process indefensible under audit review.

This design specifies that speed and scrutiny no longer compete for the same transactions: the framework concentrates Finance's desired speed on the low-risk majority, and concentrates

Compliance's desired rigor on the genuine minority — without either stakeholder's concern being deprioritized to satisfy the other.

A deliberately disclosed limitation in the design. The 0.05% tolerance baseline in FR-002 is stated explicitly as a pragmatic starting parameter intended to absorb routine FX rounding and settlement-timing variance — it is not derived from a regulatory standard or industry benchmark, and the FRD does not claim otherwise. FR-009 requires dual sign-off from both Finance and Compliance before this baseline can be changed, specifically because neither stakeholder should be able to unilaterally loosen or tighten the line between "auto-resolved" and "flagged exception." Because no mock dataset exists in this project, this baseline is never tested against real transaction variance — it remains a stated, defensible assumption rather than a validated figure, and a real implementation of this design would need to tune it against actual settlement data before go-live.

Why this approach, not a different one. An alternative design might have applied a single, moderate level of friction to every transaction — faster than the current manual process, but still requiring some review of everything. This was deliberately rejected: it would have continued to apply Compliance-level scrutiny to transactions with effectively zero risk (eroding Finance's throughput goal for no audit benefit) while diluting genuine investigative attention across a larger, less-prioritized queue (eroding Compliance's actual goal of catching real breaks). Segmenting by risk category, rather than applying uniform moderate friction, is the design choice that serves both stakeholders' real underlying interests rather than splitting the difference between their stated positions.